

Software Requirements Specification

for

Cithara

Version 1.0 approved

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Revision History

Name Date Reason For Changes Version			
Name	Date	Reason	Version
Piyawat Wiriyothin	6/2/2026	First initial SRS	1.0.0
Piyawat Wiriyayothin	9/2/2926	Correction	1.0.1

1. Introduction

1.1 Purpose

This Software Requirements Specification(SRS) document provides a comprehensive description of the Music Ai generator web application name Cithara

The purpose of this document is to formally specify the system's functionality, requirements and process of development

1.2 Document Conventions

Functional requirements are written as FR-XX

User Story are written as UR-XX

Features are written as F-XX

1.3 Intended Audience and Reading Suggestions

1. *Developer*
2. *Project Owner*
3. *Tester*
4. *Documentation writers*

1.4 Product Scope

Cithara is a web application design for the casual people who want to create their song with out musical skill .the web application work by let user generate their song using a Ai by insert a story of the song with out cost , this application will serve user as a small social which they can share their own song for the other people

1.5 References

1. *IEEE 29148 Requirements Engineering Standard*
2. *Annex-A-Detailed-Software-Requirements-Specification-SRS*

2. Overall Description

2.1 Product Perspective

The Cithara is a web application designed to help user generate song based on their prompt and selections(eg. Vocal,mood,Genre)

The system provide external services :

- *Google Oauth*
- *AI generation tool/service*

2.2 Product Functions

At a high level The Cithara provide following function

- 1. Allow users to register and login using google auth*
- 2. Allow log-in users to input a prompt to generate a song*
- 3. Require user to select a genre , vocals ,mood ,tone before generate song*
- 4. Allow user to share their generated song*
- 5. Allow users to view their generated song history*
- 6. Allow users to download their generated song*

2.3 User Classes and Characteristics

- *Creator - Primary User*
 - 1. Generated song by prompt*
 - 2. share a song*
- *Listener - Secondary User*
 - 1. View shared song*
 - 2. Can download song*

2.4 Operating Environment

- *Client Platform: Modern web browsers (e.g., Chrome, Edge, Safari) on desktop and mobile devices.*
- *Server Platform: Cloud or on-premise server environment running Windows.*
- *Backend Runtime: Node.js.*
- *Database: MySQL for storing user accounts and song metadata.*
- *External Services: Google OAuth for authentication*

2.5 Design and Implementation Constraints

The development of Cithara is constrained by the following factors:

Technical Constraints

- The system must use Google OAuth 2.0 for user authentication.
- The system must enforce access control so that only logged-in users can generate and download songs.
- The system must support files for downloads (e.g., MP3, M4A), depending on the generation provider.

Operational Constraints

- Free to prompt generate with ai
- The system must generate song only the song provided through a design form

Regulatory and Policy Constraints

- The system shall applicable with data privacy and user information protection rules

2.6 User Documentation

The system will provide

- *An easy step by step guide line for new user*
- *Inline helper in every text input box*
- *Show a feedback , error message in user interface*
- *Show estimated time while generate and progress bar*

2.7 Assumptions and Dependencies

The following assumptions and dependencies :

- *The AI music generation service is available*
- *Google authentication services are available*
- *The server is running*
- *Users have a stable internet connection*
- *The system is run every system condition correctly*

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a web-based user interface for song generation and management

- Login/Register page
- User logout
- User browse generated music
- Song preview
- display consistent navigation and error messages across all screens

3.2 Hardware Interfaces

The product is intended to run on standard consumer devices web browser. No direct hardware integration is required

- Audio (e.g., Speaker)
- Storage to download the song
- standard input device (e.g., keyboard)

3.3 Software Interfaces

The system interfaces with external software components including:

- Google OAuth 2.0 for authentication
- a relational database for storing user accounts and song metadata
- song generation service API for producing audio outputs.
- The backend exposes REST APIs to the frontend for operations such as generate, regenerate, list songs, preview, share, and download. Input data includes user prompts and configuration selections (genre, occasion, vocals)

3.4 Communications Interfaces

All client-server communication shall use HTTPS over HTTP/1.1 or HTTP/2. API requests and responses shall be formatted as JSON. The system shall protect sensitive data in transit using TLS and enforce authentication for restricted operations

4.Stakeholders

All stakeholders involved in the Cithara

Stakeholder	Role
User	Primary user of system who generate the song ,listen' and share
Project Owner	The client who defines requirement of system , scope ,and big picture of this project objective
AI generative	The external system responsible for generate song
Google Auth	External third party authentication to secure user login
Developer	Designing system , solve client solution

5. User Stories

5.1 Login/register

- **US1:**As a user, I want to sign up and log in so that I can access my account and save my generation history.
- **US2:**As a user, I want clear labels and helper text for inputs so that I know what information to fill in.
- **US20:**As a user, I want a logout page so that I can logout from my account.

5.2 Song Creation

- **US3:**As a user, I want usage guidelines so that I know how to use the website correctly.
- **US4:**As a user, I want a prompt text box so that I can describe how I want my song to be generated.
- **US5:**As a user, I want to choose the genre so that I can generate a song in my desired genre.
- **US6:**As a user, I want to select a voice and tone so that I can control the vocal style and overall mood of my song.
- **US7:**As a user, I want to edit the song's story so that I can generate a song based on my own narrative.
- **US8:**As a user, I want to set the intended event for the song so that I can organize songs and generate music tailored to that event.
- **US9:**As a user, I want a title field so that I can name my song
- **US14:**As a user, I want a regenerate option so that I can regenerate my song.
- **US15:**As a user, I want to regenerate with the same settings so that I can explore different variations quickly.

5.3 Browsing and Listening

- **US10:**As a user, I want a progress bar so that I can track the generation progress.
- **US11:**As a user, I want feedback or error reports so that I can understand what went wrong with my generated song.
- **US12:**As a user, I want to preview the generated song so that I can decide whether to keep it or regenerate it.
- **US13:**As a user, I want a media controller bar so that I can control music playback
- **US16:**As a user, I want a song browser/library so that I can view all my songs.

5.4 Sharing

- **US17:**As a user, I want a download button so that I can download the music I generated.
- **US18:**As a user, I want to choose the file format so that I can download the song in the format that I want (e.g., MP3/M4A).
- **US19:**As a user, I want a share button so that I can share my music with others.

6. System Features

this high level feature of the system that base on user stories

6.1 System Feature 1 : Register & Login

Feature ID:F1

Description:

The system provide user register and login via google authentication to ensure that all generated song is generated by a login user and all the accessibility of web application is support only logged in user

Key Capabilities:

- Email and password registration
- Google Oauth for authentication
- Access only authenticated users only

Priority:High - Essential for system

RelatedUserStories:US1,US2,US20

Related Functional Requirements:FR1,FR2,FR3,FR4,FR5,FR26

6.2 System Feature 2 : SongCreation

Feature ID:F2

Description:

The system provides a text , guideline for user step by step guide user how to create music and in each input box have a hint text what user need to input and create the song and user can regenerate the song

Key Capabilities:

- Step by Step of song creation
- Song story
- Song title
- Singer Vocal
- Mood
- Genre
- Occasion

Priority:High - Core functionality

RelatedUserStories:US3,US4,US5,US6,US7,US8,US9

RelatedFunctionalRequirements:FR6,FR7,FR8,FR9,FR10,FR11,FR12,FR13,FR14,FR15,FR16,FR17FR24,FR25

6.3 System Feature 3 : Browse Music Player

Feature ID:F3

Description:

The system provides a browse of the owner's song and library for the created song history and has a music controller for playback of each song.

Key Capabilities:

- Song playback Controller
- Show all user's generated song
- Detailed all song in library

Priority:High - Core functionality

RelatedUserStories:US13,US14,US15,US16

Related FunctionalRequirements:FR18,FR19,FR27,FR28,FR29

6.4 System Feature 4 : Sharing

Feature ID:F4

Description:

The system shall allow other user view & listen other user shared song by a link and allow user to download a song

Key Capabilities:

- Shared song
- Other user listened a shared song
- User can download a song as design file type

Priority:High - Core functionality

RelatedUserStories:US17,US18,US19,US20

Related FunctionalRequirements:FR20,FR21,FR22,FR23

7. Use Case Modeling

This section explain the flow of the system

7.1 Use Cases Diagram

Actor

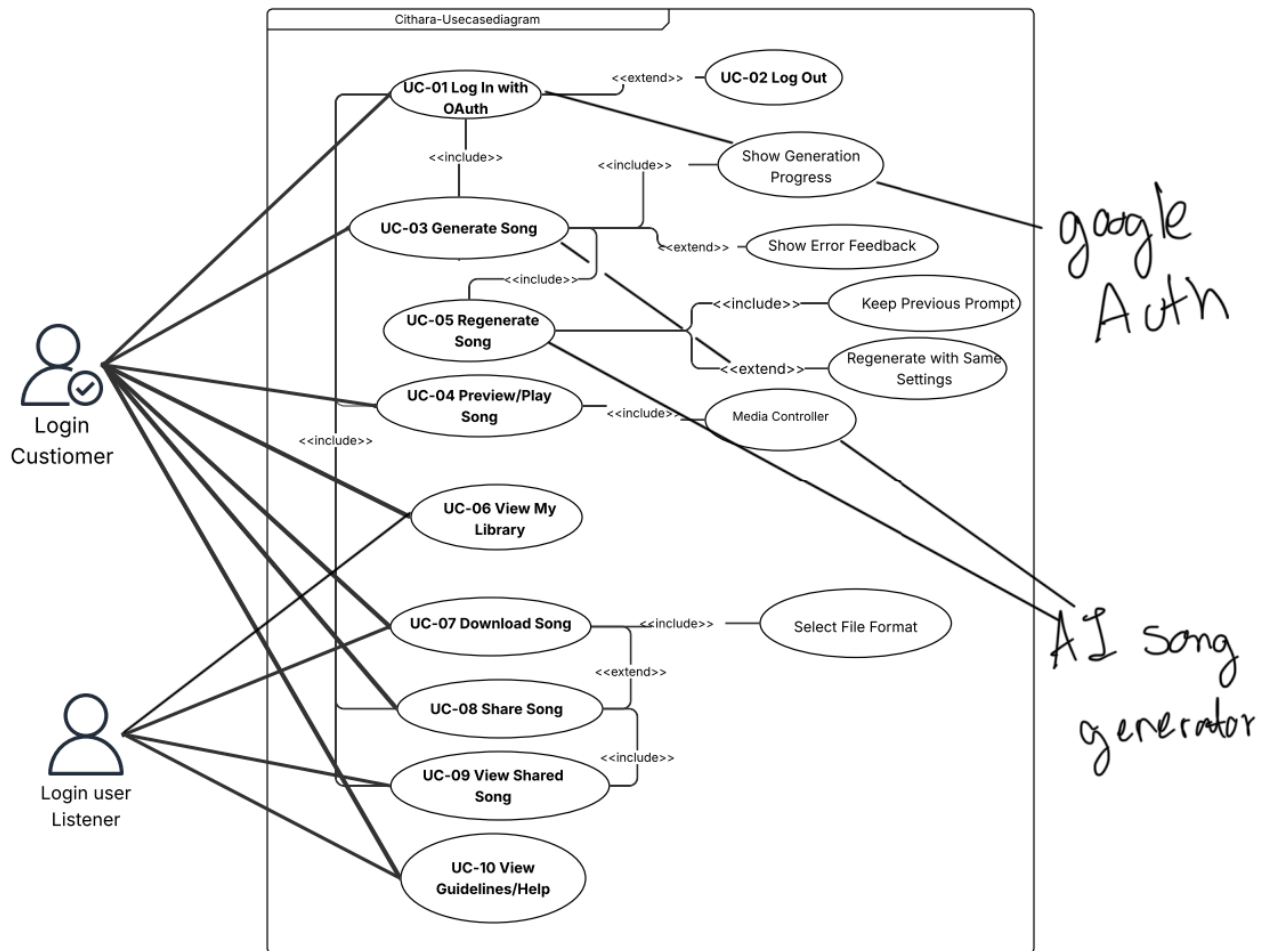
- **User (Authenticated User):** a logged-in user who can generate, manage, and share songs

Secondary Actors

- OAuth Provider (e.g., Google OAuth): authenticates users and returns tokens.
- AI Music Generation Service: generates songs from user inputs.

Use Cases:

- **UC-01 Log In with OAuth**
- **UC-02 Log Out**
- **UC-03 Generate Song**
- **UC-04 Preview/Play Song**
- **UC-05 Regenerate Song**
- **UC-06 View My Library**
- **UC-07 Download Song**
- **UC-08 Share Song**
- **UC-09 View Shared Song**
- **UC-10 View Guidelines/Help**



7.2 Use Case Modeling

7.2.1 UC-01 Log In with OAuth

- **Primary Actor:** Guest
- **Secondary Actor:** OAuth Provider
- **Goal:** Authenticate via OAuth to access core features
- **Preconditions:** User is not logged in
- **Trigger:** User clicks “Log in with OAuth”
- **Main Flow:**
 - User clicks “Log in with OAuth”.
 - System redirects the user to the OAuth Provider.
 - User grants permission on the provider page.
 - OAuth Provider returns an authorization code/token.
 - System validates the token and creates a session.
 - System redirects the user to the main page (e.g., Generate or Library).
- **Postconditions:** User becomes authenticated
- **Exceptions:**
 - User cancels / token invalid → system shows an error and returns to login.

7.2.2 UC-02 Log Out

- **Primary Actor:** User
- **Goal:** End the session and return to Guest state
- **Preconditions:** User is logged in
- **Main Flow:**
 1. User clicks “Log out”.
 2. System clears session/token.
 3. System redirects to the login page.
- **Postconditions:** User is logged out

7.2.3 UC-03 Generate Song

- **Primary Actor:** User
- **Secondary Actor:** AI Music Generation Service
- **Goal:** Generate a song using prompt and settings
- **Preconditions:** User is logged in
- **Trigger:** User clicks “Generate”
- **Main Flow:**
 - User enters a Prompt.
 - User selects a Genre (required).
 - User selects an Occasion/Event.
 - User enters a Theme.
 - User enters a Story (required).
 - User selects Voice/Tone/Vocal settings.
 - User sets a Title.
 - System validates inputs and enforces max duration = 15 minutes.
 - System sends the request to the AI Generation Service.
 - System displays a progress indicator during generation.
 - On success, system stores the generated song + metadata and opens the preview.
- **Postconditions:** A new generated song is saved in the user’s library
- **Exceptions/Extensions:**
 - Missing required fields → system blocks generation and shows validation errors.
 - Generation fails → system shows error feedback and suggests next steps.

7.2.4 UC-04 Preview/Play Song

- **Primary Actor:** User
- **Goal:** Listen to a generated song
- **Preconditions:** A song exists (from generation or library)
- **Main Flow:**
 1. User opens a song and clicks Preview/Play.
 2. System loads/streams the audio.
 3. System shows media controls (play/pause/seek/volume).
 4. System shows playback time (current time / total duration).
- **Postconditions:** Song is played for the user

7.2.5 UC-05 Regenerate Song

- **Primary Actor:** User
- **Goal:** Generate a new variation of a song
- **Preconditions:** User has a previous prompt/settings (or an existing song)
- **Main Flow:**
 1. User clicks Regenerate.
 2. System keeps the previous prompt.
 3. User chooses to regenerate with the same settings or edits inputs.
 4. System generates a new version and shows progress.
 5. System saves the new version to the library.
- **Postconditions:** A new variation is saved

7.2.6 UC-06 View My Library

- **Primary Actor:** User
- **Goal:** View all generated songs
- **Preconditions:** User is logged in
- **Main Flow:**
 1. User opens the Library page.
 2. System lists all songs generated by the user.
 3. User selects a song to preview/download/share.
- **Postconditions:** User can manage songs from the library

7.2.7 UC-07 Download Song

- **Primary Actor:** User
- **Goal:** Download a generated song
- **Preconditions:** User owns the song
- **Main Flow:**
 1. User clicks Download.
 2. System asks the user to select a file format (e.g., MP3/M4A).
 3. System prepares the file and starts the download.
- **Postconditions:** Song file is downloaded

7.2.8 UC-08 Share Song

- **Primary Actor:** User
- **Goal:** Share a generated song with others
- **Preconditions:** User owns the song
- **Main Flow:**
 1. User clicks Share.
 2. System generates a shareable link.
 3. User copies and sends the link.
- **Postconditions:** A share link exists for the song

7.2.9 UC-09 View Shared Song

- **Primary Actor:** Guest/User
- **Goal:** View a song shared by someone else
- **Preconditions:** A share link exists
- **Main Flow (login is required):**
 - User opens the shared link.
 - System checks authentication status.
 - If not logged in, system redirects to OAuth login.
 - After login, system redirects back to the shared song page.
 - User can preview/play the shared song.
- **Postconditions:** Shared song is accessible based on permissions
- **Alternative (optional design):**
 - Guests can view public shared songs without login.

7.2.10 UC-10 View Guidelines/Help

- **Primary Actor:** Guest/User
- **Goal:** Understand how to use the website
- **Main Flow:**
 1. User opens the Help/Guidelines page or sees tooltips/hints.
 2. System displays usage instructions.
- **Postconditions:** User understands the workflow

7.3 Use Case Traceability Matrix

Use Case	Related User Stories	Related FRs
UC-01 Log In with OAuth	US-01	FR-01, FR-02, FR-03
UC-02 Log Out	US-02	
UC-03 Generate Song	US-04, US-05, US-06, US-07, US-08, US-09, US-10, US-11, US-02	FR-02, FR-03, FR-05, FR-06, FR-07, FR-08, FR-09, FR-10, FR-11, FR-12, FR-13, FR-14, FR-23, FR-24
UC-04 Preview/Play Song	US-12, US-13	FR-02, FR-03, FR-18, FR-25, FR-26, FR-27
UC-05 Regenerate Song	US-14, US-15, US-10, US-11	FR-02, FR-03, FR-12, FR-13, FR-14, FR-15, FR-16
UC-06 View My Library	US-16	FR-02, FR-03, FR-17
UC-07 Download Song	US-17, US-18	FR-02, FR-03, FR-21, FR-22
UC-08 Share Song	US-19	FR-02, FR-03, FR-20
UC-09 View Shared Song		FR-04, FR-19
UC-10 View Guidelines/Help	US-03, US-02	

8.Functional Requirement

The following functional requirements for feature and usecase :

8.1 Authentication & Account Access

- **FR-01:** The system shall allow users to sign up and log in.
- **FR-02:** The system shall allow users to register and log in using Google authentication (OAuth).
- **FR-03:** The system shall restrict access to authenticated users only (non-authenticated users shall not be allowed to access protected features).
- **FR-04:** The system shall allow authenticated users to access all core features (generate, regenerate, preview, download, share).
- **FR-05:** The system shall allow users to log out from their account.
- **FR-06:** The system shall require users to log in before viewing shared songs.

8.2 Song Creation Inputs (Before Generation)

- **FR-07:** The system shall allow users to enter a prompt to generate a song.
- **FR-08:** The system shall require genre selection before generating a song.
- **FR-09:** The system shall allow users to select the intended occasion/event for the song before generating a song.
- **FR-10:** The system shall allow users to enter a song theme before generating a song.
- **FR-11:** The system shall require users to enter a song story before generating a song.
- **FR-12:** The system shall allow users to select vocal/singer settings before generating a song.
- **FR-13:** The system shall allow users to set a title for their songs.
- **FR-14:** The system shall limit the maximum generated song duration to 15 minutes.

8.3 Song Generation Process & Error Handling

- **FR-15:** The system shall display generation progress while a song is being generated.
- **FR-16:** The system shall display an error message when generation fails.
- **FR-17:** The system shall provide actionable feedback when an error occurs.

8.4 Regeneration (Variations)

- **FR-18:** The system shall allow users to regenerate a song.
- **FR-19:** The system shall keep the previous prompt when users click regenerate.

8.5 Library & Viewing

- **FR-20:** The system shall display a list of all songs generated by the authenticated user.
- **FR-21:** The system shall allow users to preview generated songs.
- **FR-22:** The system shall allow users to view songs shared by other users (friends).

8.6 Sharing

- **FR-23:** The system shall allow users to share their generated songs.

8.7 Download

- **FR-24:** The system shall allow users to download their own generated songs.
- **FR-25:** The system shall allow users to select a download file format (e.g., MP3/M4A).

8.8 Playback (Media Controls)

- **FR-26:** The system shall provide a media control bar for song playback.
- **FR-27:** The system shall display playback controls while a song is playing.
- **FR-28:** The system shall display the playback duration (current time / total duration) during playback.

8.9 Usability (UI Help)

- **FR-29:** The system shall display a tooltip or hint in the prompt text box.

9. Other Nonfunctional Requirements

9.1 Performance Requirements

- *The system shall show a visible loading/progress indicator during song generation and regeneration.*
- *The system should keep the UI responsive while generation is running*

9.2 Security Requirements

- *The system shall use HTTPS for all pages and API requests.*
- *The system shall enforce authentication rules: only logged-in users can access protected features and view shared songs*
- *The system shall ensure users can download and manage only their own generated songs, unless a song is shared with them.*

9.3 Usability Requirements

- *The system shall provide clear labels and helper text for key inputs (prompt, genre, story, voice/tone)*
- *The system shall provide basic usage guidelines (Help/Guide) that explain how to generate, preview, download, and share songs.*